

PROTEIN STRUCTURE CLASSIFICATION

USING MACHINE LEARNING

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INTRODUCTION

A. Brief Overview of the Protein Classification Project

- **Objective** | The primary aim of this project is to test a machine-learning approach for the classification of proteins based on their structural attributes and experimental characteristics.
- **Dataset** | We are working with a comprehensive dataset that includes information on protein structures, experimental techniques, and other relevant features.
- **Classification Target** | The 'classification' column serves as our target variable, representing different classes or functional categories of proteins.
- **Machine Learning Approach** | Leveraging machine learning for protein classification enables the development of predictive models that can enhance our ability to discern functional patterns from complex biological data

B. Importance of Understanding Protein Structure:

- **Significance** | Understanding the relationship between a protein's structure and its function is fundamental to unraveling the mysteries of biology.
- **Functional Diversity:** | Proteins play diverse roles in living organisms, from catalyzing biochemical reactions to serving as structural components. The structure-function relationship is central to deciphering these roles.
- **Biomedical Implications:** | Insights into protein structure-function relationships have profound implications for drug discovery, disease understanding, and the design of therapeutic interventions.
- **Importance of Classification:** | Accurate classification of proteins allows us to categorize them based on shared characteristics, facilitating targeted research and advancing our understanding of biological systems.

DATA SET OVERVIEW

This is a protein data set obtained from Kaggle which was retrieved from the Research Collaboratory for Structural Bioinformatics (RCSB) Protein Data Bank (PDB)

A. Data Source

This is a protein data set obtained from Kaggle which was retrieved from the Research Collaboratory for Structural Bioinformatics (RCSB) Protein Data Bank (PDB)

B. Data Contents

- There are two data files. Both are arranged on the "structureId" of the protein. pdb_data_no_dups.csv contains protein metadata which includes details on protein classification, extraction methods, etc. data_seq.csv contains >400,000 protein structure sequences.
- **Number of Records:** 111332 data points (after data cleaning).
- **Number of Columns:** 16 features

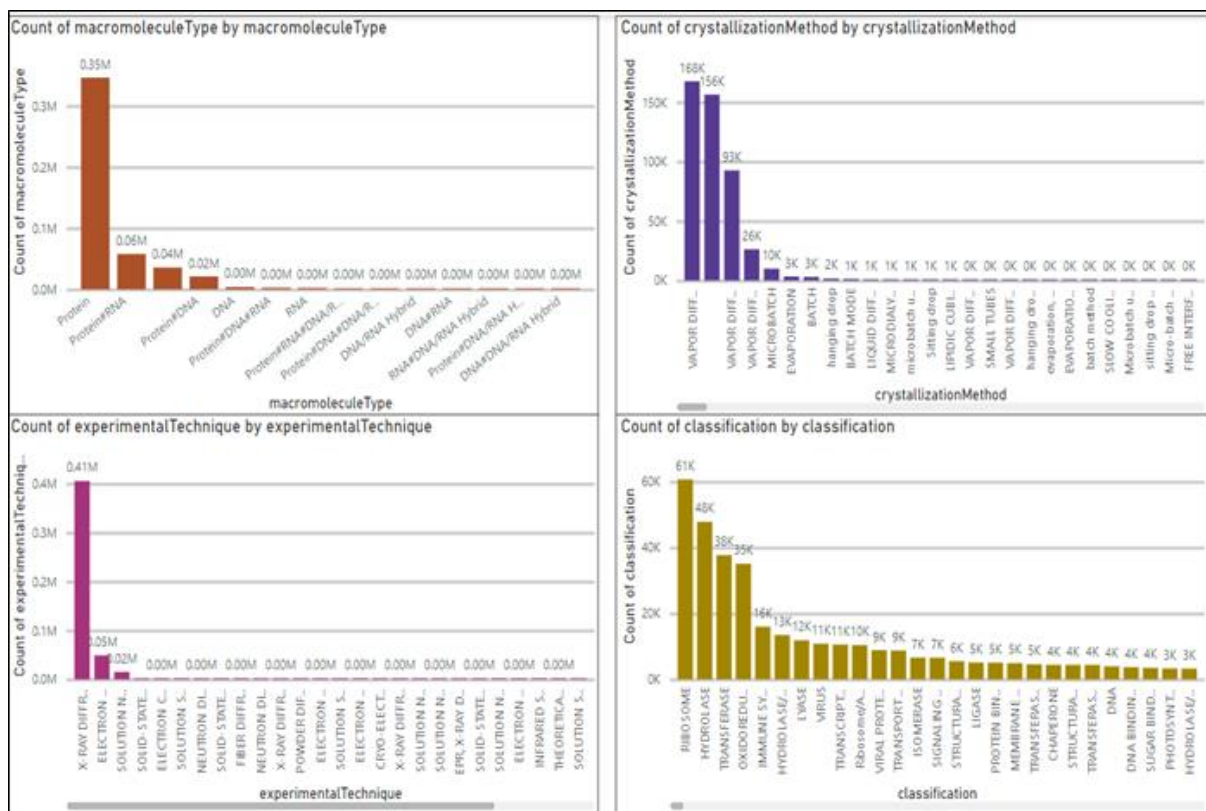
C. Feature Description

<u>Sr.No.</u>	<u>Column Name</u>	<u>Description</u>
1.	structureId	This column represents a unique identifier for each protein structure in the dataset
2.	classification	This column contains information about the classification category to which the protein structure belongs.
3.	experimentalTechnique	This column contains information about the experimental technique used to determine the protein structure (X-Ray Diffraction, Electron Microscope, etc).
4.	macromoleculeType	This column likely describes the type of macromolecule (Protein, DNA/RNA, nucleic acid, etc)
5.	residueCount	This column represents the total number of amino acid residues in the protein structure.
6.	resolution	This column is associated with the resolution of the experimental technique used to determine the protein structure. It indicates the level of detail in the structure.
7.	structureMolecularWeight	This column contains information about the molecular weight of the protein structure, indicating the mass of the entire structure.
8.	crystallizationMethod	This column contains details about the method used for crystallizing the protein before structure determination.
9.	crystallizationTemperatureK	This column represents the temperature at which crystallization occurred, measured in Kelvin.
10.	densityMatthews	This column contains information about the Matthews coefficient, which is a parameter used in crystallography to estimate the solvent content in the crystal.
11.	densityPercentSol	This column contains the percentage of solvent in the crystal, providing additional information about the crystal packing.
12.	pdbxDetails	This column contains additional details or comments related to the Protein Data Bank (PDB) file, which is a database for the three-dimensional structural data of large biological molecules.
13.	pHValue	This column contains information about the pH value of the experimental conditions during the structure determination process.

14.	publicationYear	This column represents the year in which the corresponding research or study was published.
15.	chainId:	This column likely contains information about the chain identifier associated with the protein structure.
16.	sequence:	This column probably contains the primary sequence of the macromolecule

Exploratory Data Analysis (EDA)

A. EDA on Raw Data



Findings From the Above Graph

1. The most common macromolecule type is protein, with a count of over 250,000
2. The most common crystallization method is vapor diffusion, with a count of over 150,000
3. The most common experimental technique is X-ray diffraction, with a count of over 250,000
4. The most common classification method is alpha and beta proteins, with a count of over 100,000
5. The Protein and Protein#RNA are the macromolecules that contribute more than 92.8% of the data.
6. The Ribosomes, Hydrolase, Transferase, and Oxidoreductase are the protein classes that contribute a large portion of the data. All other classes have a smaller number of observations.

B. Data Preparation and Data Cleaning

1. From the above findings we decided to filter the data to the top 2 macromolecules which are Protein and Protein#RNA as they contribute more information to the data set.
2. All other classes that are contributing a smaller number of observations are eliminated, and the top 3 classes are used for further data analysis namely Hydrolase, Transferase, and Oxidoreductase. Although Ribosomes contribute a large amount of information, based on domain knowledge we found that Ribosome can easily be differentiated because of their properties.
3. We only chose the X-ray diffraction technique as it is mostly used experimental technique and all other methods are excluded.
4. After performing all the above steps, we dropped all null values in the filtered data set.
5. After filtering and dropping the null values we set the 'structureId' column as the index of the data frame.
6. The target column 'classification' is encoded by using label encoding.
7. Based on domain knowledge, we dropped all other categorical columns as they contain fact-based information which is not going to affect the class of the protein

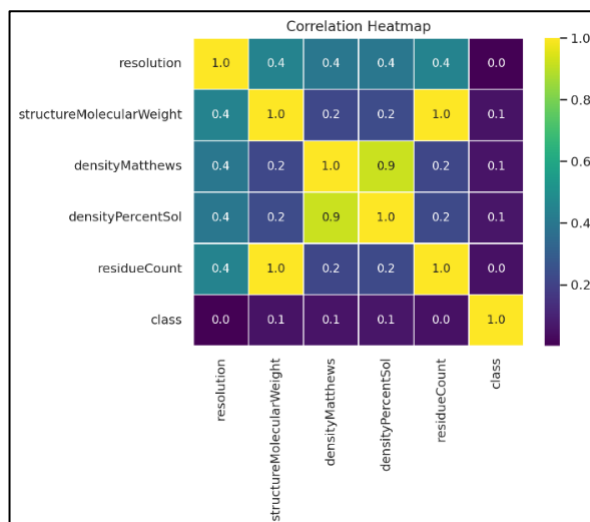
C. Statistical Analysis

	resolution	structureMolecularWeight	densityMatthews	densityPercentSol	residueCount	class
COUNT	111332.00	111332.00	111332.00	111332.00	111332.00	111332.00
MEAN	2.23	183466.00	2.69	52.00	1630.61	1.91
STD	0.59	246061.80	0.65	9.26	2238.30	0.82
MIN	0.64	2379.71	0.97	0.28	20.00	1.00
25%	1.85	54941.59	2.27	45.70	484.00	1.00
50%	2.15	105845.20	2.51	51.00	932.00	2.00
75%	2.52	214043.50	2.91	57.62	1900.00	3.00
MAX	9.01	4918807.00	13.89	88.00	57792.00	3.00

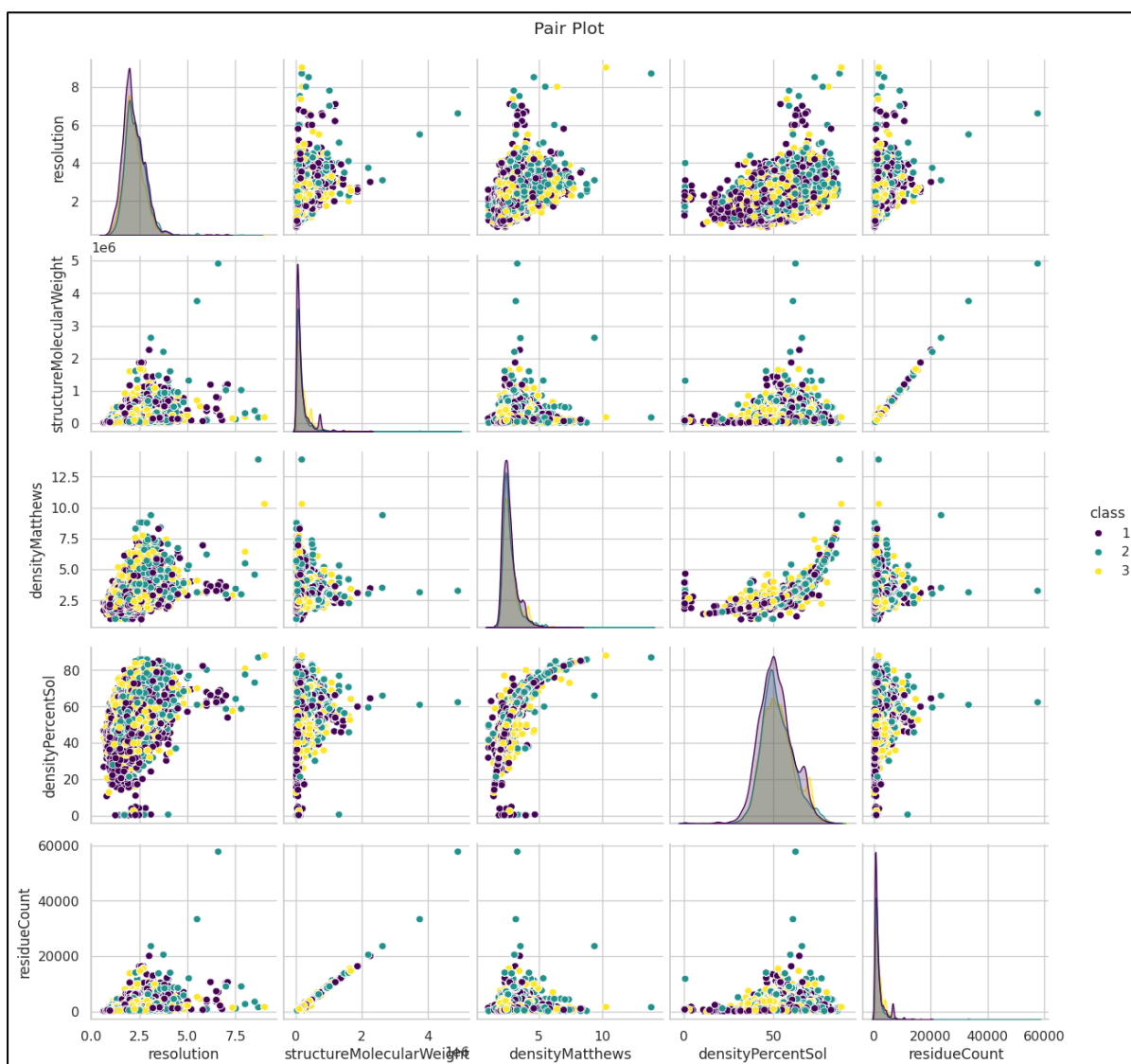
D. Correlation Matrix:

	resolution	structureMolecularWeight	densityMatthews	densityPercentSol	residueCount	class
resolution	1.00	0.45	0.40	0.40	0.45	0.00
structureMolecularWeight	0.45	1.00	0.22	0.23	1.00	0.05
densityMatthews	0.40	0.22	1.00	0.92	0.21	0.05
densityPercentSol	0.40	0.23	0.92	1.00	0.22	0.06
residueCount	0.45	1.00	0.21	0.22	1.00	0.05
class	0.00	0.05	0.05	0.06	0.05	1.00

E. Correlation Matrix Heatmap



F. Pair Plot



MODEL BUILDING

In this section, we detail the construction, evaluation, and comparative analysis of two key machine-learning models for protein classification: the K-Nearest Neighbors (KNN) classifier and the Random Forest classifier. Both models were trained on a carefully pre-processed dataset of proteins, and their performances were evaluated using classification reports and confusion matrices.

A. K-Nearest Neighbors (KNN) Classifier

1. Model Development

- **Objective:**

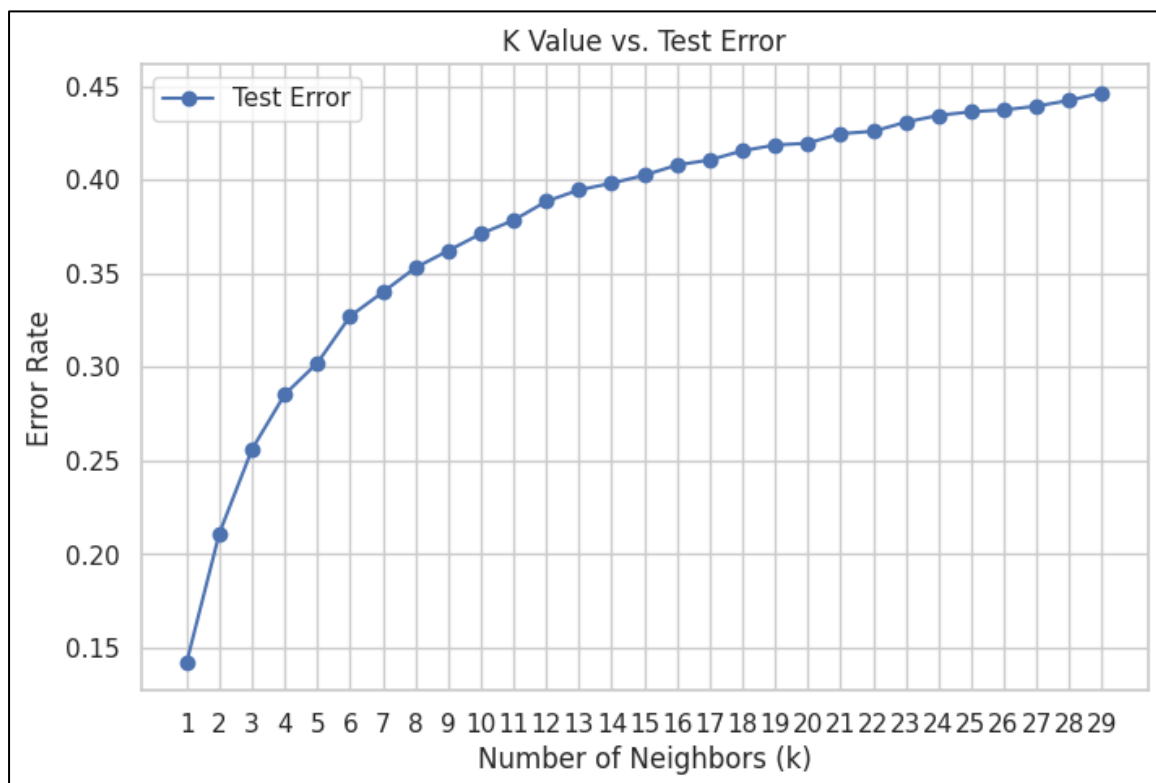
The KNN model was implemented to classify proteins based on numerical features derived from structural and experimental data.

- **Preprocessing:**

The feature set was standardized using StandardScaler to ensure that all features contributed equally to the Euclidean distance calculations used in KNN.

- **Hyperparameter Tuning:**

Using the elbow method, we examined different values of k . Our experiments showed that the model achieved the lowest test error when $k = 1$, indicating that considering only the closest neighbor resulted in optimal performance.



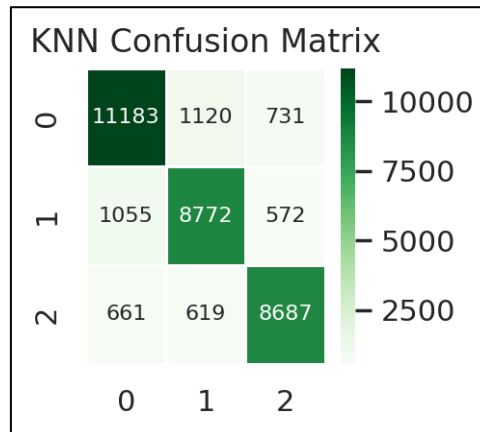
2. Results and Interpretation

- **Performance Metrics:**

The final KNN model with $k = 1$ achieved competitive accuracy, as demonstrated by the detailed classification report. The confusion matrix further illustrated that misclassifications were minimal, confirming the model's effectiveness in distinguishing between the three target protein classes.

- **Visualization:**

A confusion matrix plot was generated to provide a visual summary of the model's performance.



B. Random Forest Classifier

1. Model Development

- **Objective**

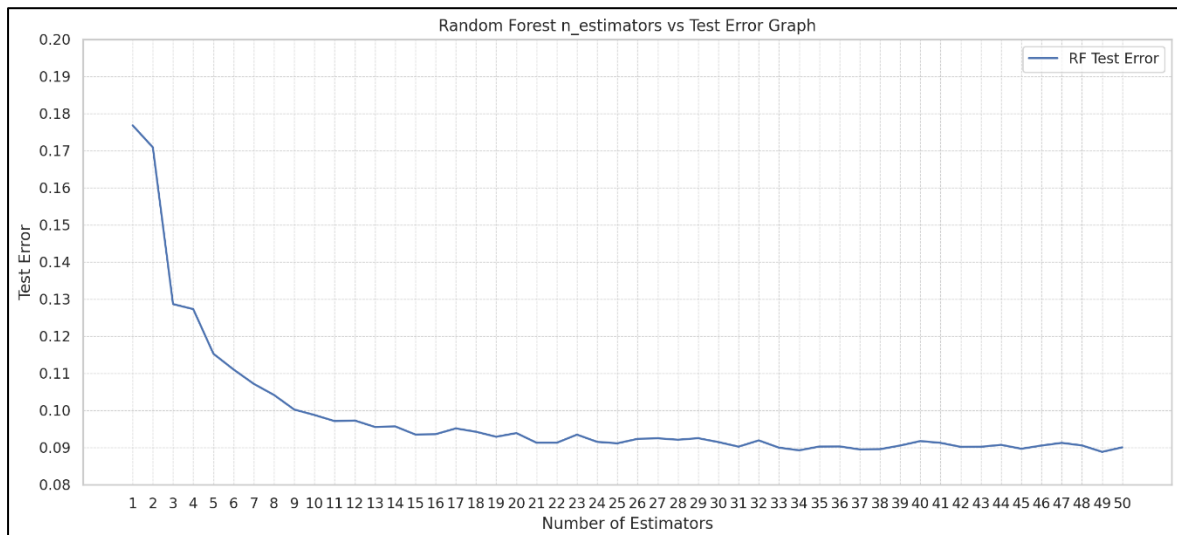
The Random Forest model was developed to improve predictive accuracy by combining the outputs of multiple decision trees trained on different sub-samples of the data.

- **Preprocessing**

As with KNN, the features were standardized to ensure consistent scaling across all variables.

- **Hyperparameter Tuning with GridSearchCV**

Below is the graph of for estimating the optimal value of `n_estimators`:



To optimize the performance of the Random Forest model, we employed GridSearchCV for hyperparameter tuning. In this process, we explored various configurations for the number of estimators (`n_estimators`), maximum tree depth, and other parameters. The grid search helped identify the optimal combination that minimized the test error.

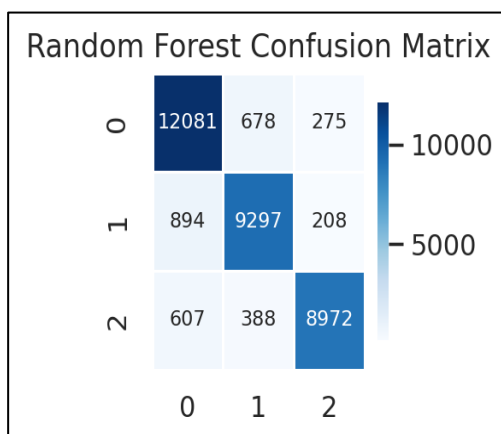
2. Results and Interpretation

- **Performance Metrics**

The tuned Random Forest model demonstrated robust performance with balanced precision, recall, and F1 scores across the three protein classes. The confusion matrix further validated the model's strong classification ability.

- **Visualization:**

A confusion matrix plot was generated for the Random Forest model, and a graph of test error vs. number of estimators was used to visualize the hyperparameter tuning results.



C. Comparative Analysis and Performance Summary

Both the KNN and Random Forest models have demonstrated promising results in classifying proteins. The KNN model, with its simplicity ($k = 1$), provided a strong baseline, while the Random Forest model, enhanced through rigorous hyperparameter tuning with GridSearchCV, offered a robust and more stable performance overall.

Summary Table of Performance Metrics:

Below is a suggested table format to summarize the performance metrics (accuracy, precision, recall, and F1-score) for both models. Insert this table into your report after the section comparing the models

Model	Protein Class	Precision	Recall	f1-score	Accuracy
KNN Classifier	HYDROLASE	0.87	0.86	0.86	0.86
	TRANSFERASE	0.83	0.84	0.84	
	OXIDOREDUCTASE	0.87	0.87	0.87	
Random Forest Classifier	HYDROLASE	0.89	0.93	0.91	0.91
	TRANSFERASE	0.9	0.89	0.9	
	OXIDOREDUCTASE	0.95	0.9	0.92	

CONCLUSION

The model-building process demonstrates that machine learning methods can effectively classify proteins into functional categories using structural and experimental data. While the KNN model provided a simple yet effective solution, the Random Forest model—with its ensemble approach and GridSearchCV-driven hyperparameter tuning—yielded enhanced performance and robustness. Future work will focus on expanding the classification to a broader range of protein classes and further refining the model by addressing feature multicollinearity and computational efficiency.

LIMITATIONS AND FUTURE WORK

While the current study demonstrates promising results, several limitations and avenues for future research remain:

1. **Limited Class Scope:**

The model is currently designed to predict only three protein classes. Expanding the classification to include a wider variety of protein categories could provide a more comprehensive understanding of protein functions.

2. **Dataset Scale:**

Future work should test the model on a larger dataset encompassing more diverse protein observations. This would help assess the model's scalability and its ability to generalize across different protein classes.

3. **Feature Multicollinearity:**

The current analysis does not address multicollinearity among features. Investigating and mitigating collinearity could improve model stability and interpretability, potentially leading to more robust predictions.

4. **Computational Efficiency:**

As the dataset size increases, so does the computational cost of training and tuning the models. Future research should focus on optimizing the model architecture and exploring efficient algorithms to reduce computational overhead while maintaining or improving accuracy.

These improvements could enhance the model's predictive performance and broaden its applicability in the field of protein structure classification.